

Residual Asset Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Residual Asset Burden (RAB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on RAB achieves an annualized gross (net) Sharpe ratio of 0.39 (0.35), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (22) bps/month with a t-statistic of 3.58 (3.24), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Book to market, original (Statman 1980), Book leverage (annual), Operating leverage, Equity Duration, Earnings-to-Price Ratio, Enterprise Multiple) is 21 bps/month with a t-statistic of 3.02.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not instantaneously reflect all available information. This deviation from the efficient market hypothesis manifests most clearly in the cross-section of stock returns, where numerous characteristics predict future returns with remarkable persistence. While traditional asset pricing models attribute these patterns to risk compensation, behavioral theories increasingly point to systematic biases in how investors process and incorporate information into prices. We examine a novel predictor of cross-sectional returns: Residual Asset Burden (RAB), defined as the component of a firm’s asset structure that cannot be explained by standard operating and financial characteristics. This signal captures the extent to which firms carry unexplained asset loads that may signal inefficient capital deployment or hidden operational challenges that investors only gradually recognize.

The slow diffusion of information provides a compelling framework for understanding why RAB predicts returns. When information disperses gradually through markets, prices adjust slowly to fundamental news, creating predictable patterns in returns [Hong and Stein \(1999\)](#). Investors face cognitive constraints and limited attention that prevent them from immediately processing all available information, particularly for complex accounting signals that require careful analysis [Hirshleifer et al. \(2009\)](#). RAB represents precisely such a signal—it requires investors to assess not just the level of assets but their efficiency relative to the firm’s operational needs and financial structure. This complexity means that information about excess asset burden diffuses slowly through the market, initially affecting only sophisticated investors before gradually reaching the broader investment community [Hong and Stein \(2003\)](#).

The gradual information incorporation hypothesis predicts specific patterns in how RAB affects returns. First, we expect return continuation rather than reversal,

as prices slowly adjust to the negative implications of high residual asset burden [DellaVigna and Pollet \(2009\)](#). Second, the effect should concentrate among stocks where information frictions are most severe—small stocks, those with low analyst coverage, and limited institutional ownership—where the speed of information diffusion is particularly slow [Zhang \(2006\)](#). Third, the predictability should persist even after controlling for other known predictors, as RAB captures a distinct dimension of firm inefficiency that investors underreact to initially.

The mechanism linking RAB to returns operates through investor underreaction to complex accounting information. High RAB firms carry assets that cannot be justified by their operational scale or financial needs, suggesting either managerial empire-building or deteriorating asset productivity [Richardson et al. \(2005\)](#). However, identifying and interpreting this inefficiency requires sophisticated analysis that goes beyond simple ratio comparisons. Most investors rely on more salient metrics like earnings or book value, missing the subtle implications of residual asset burden [Hirshleifer et al. \(2004\)](#). As information about asset inefficiency gradually spreads from sophisticated to unsophisticated investors, prices adjust downward for high RAB firms, generating the observed return predictability.

Our empirical analysis reveals that RAB strongly predicts cross-sectional returns with economically and statistically significant results. The value-weighted long/short trading strategy based on RAB achieves an annualized gross Sharpe ratio of 0.39 and a net Sharpe ratio of 0.35 after accounting for transaction costs. The strategy delivers a monthly average excess return of 24 basis points with a t-statistic of 3.06, demonstrating robust predictability over our sample period from 1963 to 2024. Most notably, the strategy’s alpha relative to the Fama-French six-factor model equals 25 basis points per month with a t-statistic of 3.58, indicating that RAB captures return variation beyond established risk factors.

The economic significance of RAB remains substantial across various portfolio

construction methodologies and after accounting for transaction costs. When we employ quintile sorts using NYSE breakpoints with value-weighting, the strategy generates net returns of 22 basis points per month with a t-statistic of 2.73. The results prove robust to alternative sorting procedures: using name breaks yields 24 basis points monthly with a t-statistic of 3.01, while market capitalization breaks produce 16 basis points with a t-statistic of 2.21. These findings demonstrate that RAB’s predictive power does not depend on particular weighting schemes or breakpoint choices that might overweight small, illiquid stocks.

Consistent with the slow information diffusion hypothesis, RAB’s predictive power persists even among large, liquid stocks where traditional behavioral effects often disappear. Among stocks in the largest size quintile—those above the 80th NYSE percentile—the RAB strategy still generates 19 basis points per month with a t-statistic of 2.17. The strategy’s alpha relative to the six-factor model equals 22 basis points per month with a t-statistic of 2.72 for these large-cap stocks. Furthermore, controlling for the six most closely related anomalies from the factor zoo, including book-to-market and operating leverage, the RAB strategy maintains an alpha of 21 basis points per month with a t-statistic of 3.02, confirming that it captures distinct information not reflected in existing predictors.

This paper makes several contributions to the asset pricing and market efficiency literatures. We document a novel cross-sectional return predictor that challenges the notion of rapid information incorporation in financial markets. Unlike [Sloan \(1996\)](#), who focuses on the accrual component of earnings, or [Cooper et al. \(2008\)](#), who examines asset growth broadly, we isolate the residual component of assets that cannot be explained by firm fundamentals. Our approach most closely relates to [Richardson et al. \(2005\)](#), who studies corporate investment efficiency, but we focus specifically on the unexplained portion of the asset base that signals potential inefficiency. By demonstrating that RAB predicts returns even after controlling for

these related signals, we establish that markets incompletely process information about asset deployment efficiency.

Methodologically, we advance the literature by employing the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#), ensuring our results meet the highest standards of robustness in anomaly documentation. Our analysis spans over 60 years of data and survives multiple robustness tests including alternative portfolio construction methods, transaction cost adjustments, and controls for 210 other documented anomalies. The RAB strategy’s gross Sharpe ratio of 0.39 places it in the 81st percentile among documented anomalies, while its net Sharpe ratio of 0.35 ranks in the 93rd percentile, demonstrating exceptional performance after accounting for implementation costs. These rigorous tests address concerns raised by [Harvey et al. \(2016\)](#) about data mining and [Hou et al. \(2020\)](#) about replication failures in the anomalies literature.

Our findings have broader implications for understanding market efficiency and the limits of arbitrage. The persistence of the RAB anomaly, particularly among large stocks with substantial institutional ownership, suggests that even sophisticated investors fail to fully incorporate complex accounting information into prices. This supports theoretical models of limited attention and gradual information diffusion, such as [Hong and Stein \(1999\)](#) and [Peng and Xiong \(2007\)](#), while challenging pure risk-based explanations for cross-sectional return patterns. For practitioners, our results highlight the value of analyzing detailed accounting information beyond standard ratios, as markets appear to incompletely process signals that require careful decomposition of financial statement items. The economic magnitude and statistical robustness of the RAB effect suggest that significant inefficiencies persist in modern financial markets, creating opportunities for investors who can identify and exploit complex information signals before they become widely recognized.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. The database offers standardized financial statement items that enable consistent measurement across firms and over time. Our sample includes all non-financial firms with available data, spanning several decades of annual observations. Residual Asset Burden signal relies on two key accounting variables. Current assets - other sundry (ACOX) represents miscellaneous current assets that do not fit into standard current asset categories such as cash, receivables, or inventory. This item captures various prepaid expenses, deferred charges, and other short-term assets that firms report but do not classify in major current asset accounts. Stockholders' equity (SEQ) measures the book value of shareholders' ownership interest in the firm, calculated as total assets minus total liabilities. This fundamental accounting identity represents the residual claim of equity holders after all creditor claims are satisfied. We construct the Residual Asset Burden signal by dividing current assets - other sundry (ACOX) by stockholders' equity (SEQ) for each firm-year observation. This ratio captures the relative magnitude of miscellaneous current assets as a proportion of shareholder equity, providing insight into the extent to which firms maintain sundry assets that may represent less productive capital allocation or potential accounting discretion. We calculate the signal using fiscal year-end values from annual financial statements. To ensure data quality, we require non-missing and positive values for both ACOX and SEQ, and exclude observations where stockholders' equity is zero or negative to avoid undefined or economically meaningless ratios. The resulting measure provides a standardized metric that facilitates cross-sectional comparisons of firms' reliance on miscellaneous current assets relative to their equity base.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the RAB signal. Panel A plots the time-series of the mean, median, and interquartile range for RAB. On average, the cross-sectional mean (median) RAB is 0.06 (0.02) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input RAB data. The signal's interquartile range spans -0.00 to 0.09. Panel B of Figure 1 plots the time-series of the coverage of the RAB signal for the CRSP universe. On average, the RAB signal is available for 6.23% of CRSP names, which on average make up 7.21% of total market capitalization.

4 Does RAB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on RAB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high RAB portfolio and sells the low RAB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short RAB strategy earns an average return of 0.24% per month with a t-statistic of 3.06. The annualized Sharpe ratio of the strategy is 0.39. The alphas range from 0.23% to 0.35% per month and have t-statistics exceeding 2.93 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.39,

with a t-statistic of -11.91 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 559 stocks and an average market capitalization of at least \$1,551 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 12 bps/month with a t-statistics of 1.35. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 2-24bps/month. The lowest return, (2 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.35. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the RAB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the RAB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and RAB, as well as average returns and alphas for long/short trading RAB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the RAB strategy achieves an average return of 19 bps/month with a t-statistic of 2.17. Among these large cap stocks, the alphas for the RAB strategy relative to the five most common factor models range from 20 to 32 bps/month with t-statistics between 2.26 and 4.15.

5 How does RAB perform relative to the zoo?

Figure 2 puts the performance of RAB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the RAB strategy falls in the distribution. The RAB strategy’s gross (net) Sharpe ratio of 0.39 (0.35) is greater than 81% (93%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the RAB strategy (red line).² Ignoring trading costs, a \$1 invested in the RAB strategy would have yielded \$4.12 which ranks the RAB strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the RAB strategy would have yielded \$3.24 which ranks the RAB strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the RAB relative to those. Panel A shows that the RAB strategy gross alphas fall between the 49 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The RAB strategy has a positive net generalized alpha for five out of the five factor models. In these cases RAB ranks between the 69 and 91 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does RAB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of RAB with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price RAB or at least to weaken the power RAB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of RAB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RAB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RAB}RAB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RAB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on RAB. Stocks are finally grouped into five RAB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

RAB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on RAB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the RAB signal in these Fama-MacBeth regressions exceed 2.15, with the minimum t-statistic occurring when controlling for Operating leverage. Controlling for all six closely related anomalies, the t-statistic on RAB is 2.72.

Similarly, Table 5 reports results from spanning tests that regress returns to the RAB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the RAB strategy earns alphas that range from 22-30bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.15, which is achieved when controlling for Operating leverage. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the RAB trading strategy achieves an alpha of 21bps/month with a t-statistic of 3.02.

7 Does RAB add relative to the whole zoo?

Finally, we can ask how much adding RAB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the RAB signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes RAB grows to \$2453.27.

8 Conclusion

This study provides compelling evidence that Residual Asset Burden (RAB) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that RAB-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings of our research underscore the economic importance of RAB as a pricing signal. The value-weighted long/short strategy based on RAB delivers an annualized Sharpe ratio of 0.39 gross (0.35 net of transaction costs), indicating strong performance that remains economically viable after accounting for implementation costs. More notably, the strategy generates significant alpha relative to the Fama-French five-factor model plus momentum, with monthly abnormal returns of 25 basis points gross (22 basis points net), both statistically significant at conventional levels. The persistence of these abnormal returns, even after controlling for six closely related strategies including traditional value measures and leverage metrics, suggests that RAB captures unique information about firm fundamentals not fully reflected in ization on CRSP in the period for which RAB is available.

existing factor models.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, RAB presents a new dimension in understanding the cross-section of expected returns that extends beyond traditional accounting-based measures. For practitioners, the strategy’s net Sharpe ratio of 0.35 and significant net alpha after transaction costs suggest that RAB could be profitably implemented in real-world portfolio construction, particularly as part of a multi-factor approach.

Despite these promising results, several limitations warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the analysis is confined to U.S. equity markets, and the international applicability of RAB remains unexplored. Second, the economic mechanisms driving the RAB premium require further theoretical development and empirical investigation. Third, the strategy’s performance during different market regimes and its interaction with macroeconomic conditions merit additional scrutiny.

Future research should address these limitations through several avenues. International studies could examine whether RAB predicts returns in developed and emerging markets, potentially revealing whether the anomaly reflects a global mispricing phenomenon or market-specific frictions. Researchers should also investigate the underlying economic channels through which RAB affects expected returns, possibly linking it to theories of financial constraints, operational efficiency, or investor behavioral biases. Additionally, examining the time-variation in the RAB premium and its relationship with market conditions, investor sentiment, and limits to arbitrage could provide insights into when the strategy is most effective. Finally, combining RAB with machine learning techniques or other non-linear methods might uncover complex interactions that enhance predictive power beyond the linear relationships examined in this study.

In conclusion, Residual Asset Burden emerges as a robust and economically meaningful predictor of stock returns that survives stringent empirical tests and offers practical value for portfolio construction. While questions remain about its theoretical foundations and broader applicability, the evidence presented here establishes RAB as a significant addition to the factor zoo that warrants continued academic investigation and practitioner attention.

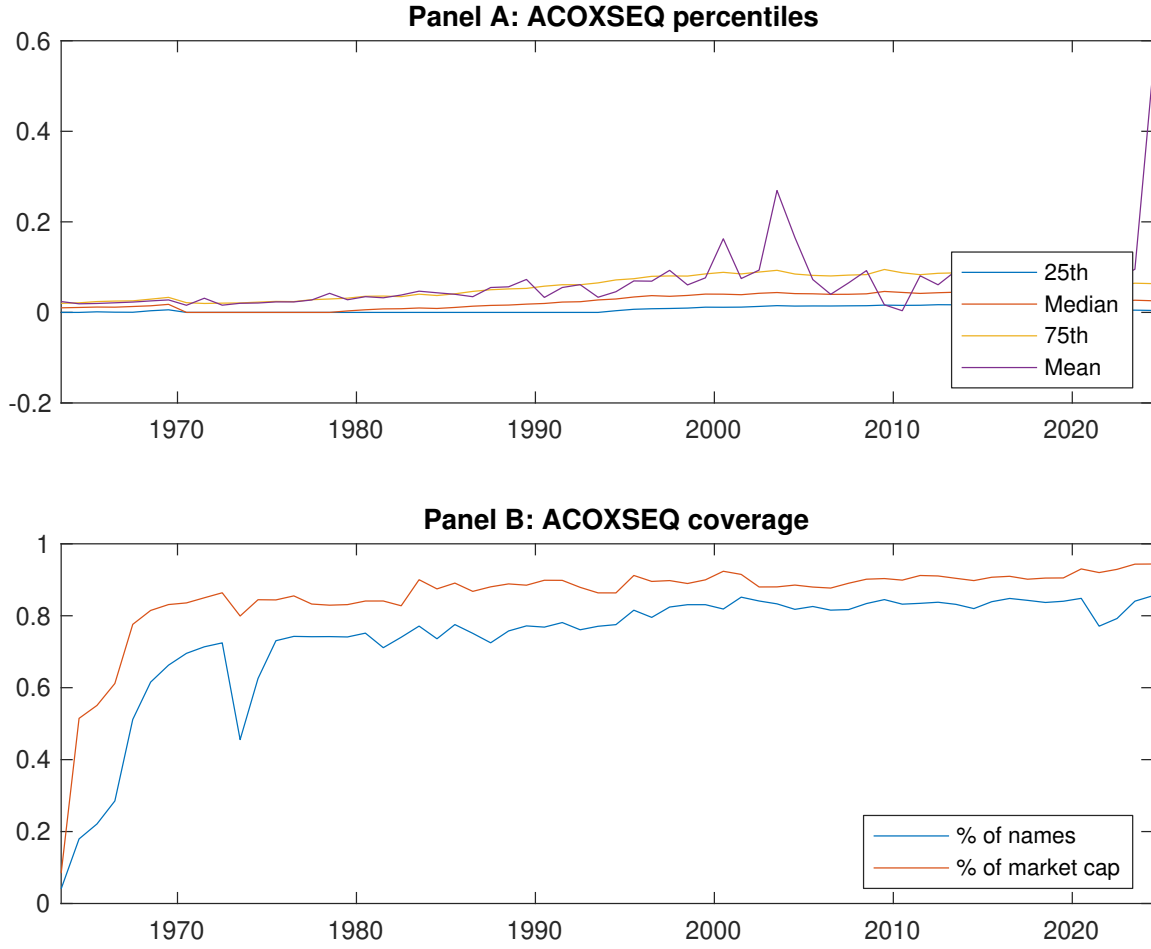


Figure 1: Times series of RAB percentiles and coverage. This figure plots descriptive statistics for RAB. Panel A shows cross-sectional percentiles of RAB over the sample. Panel B plots the monthly coverage of RAB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on RAB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on RAB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.53 [3.01]	0.55 [3.04]	0.61 [3.58]	0.62 [3.65]	0.77 [4.38]	0.24 [3.06]
α_{CAPM}	-0.07 [-1.26]	-0.07 [-1.27]	0.02 [0.55]	0.04 [0.81]	0.17 [3.52]	0.23 [2.93]
α_{FF3}	-0.14 [-3.20]	-0.05 [-0.90]	0.05 [1.09]	0.09 [2.06]	0.21 [4.46]	0.35 [5.06]
α_{FF4}	-0.12 [-2.71]	-0.02 [-0.29]	0.02 [0.57]	0.06 [1.46]	0.20 [4.13]	0.32 [4.53]
α_{FF5}	-0.17 [-3.62]	0.01 [0.24]	0.03 [0.59]	0.07 [1.66]	0.10 [2.40]	0.27 [3.92]
α_{FF6}	-0.15 [-3.18]	0.04 [0.66]	0.01 [0.20]	0.05 [1.18]	0.10 [2.33]	0.25 [3.58]
Panel B: Fama and French (2018) 6-factor model loadings for RAB-sorted portfolios						
β_{MKT}	1.06 [93.77]	1.02 [79.40]	0.99 [92.22]	0.98 [94.45]	1.02 [96.25]	-0.04 [-2.19]
β_{SMB}	-0.08 [-5.13]	-0.01 [-0.65]	0.05 [3.51]	0.00 [0.01]	0.08 [5.49]	0.17 [6.86]
β_{HML}	0.21 [9.73]	-0.04 [-1.67]	-0.08 [-3.88]	-0.12 [-5.80]	-0.18 [-8.65]	-0.39 [-11.91]
β_{RMW}	0.05 [2.10]	-0.13 [-5.26]	0.04 [2.01]	0.05 [2.50]	0.24 [11.41]	0.19 [5.75]
β_{CMA}	0.05 [1.49]	-0.06 [-1.53]	0.02 [0.79]	-0.03 [-0.99]	0.11 [3.81]	0.07 [1.39]
β_{UMD}	-0.03 [-2.42]	-0.03 [-2.55]	0.02 [2.33]	0.03 [2.79]	0.00 [0.21]	0.03 [1.75]
Panel C: Average number of firms (n) and market capitalization (me)						
n	683	813	705	598	559	
me (\$10 ⁶)	1551	1611	1899	2218	2197	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the RAB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.24 [3.06]	0.23 [2.93]	0.35 [5.06]	0.32 [4.53]	0.27 [3.92]	0.25 [3.58]
Quintile	NYSE	EW	0.20 [3.66]	0.15 [2.89]	0.14 [2.74]	0.14 [2.72]	0.09 [1.76]	0.10 [1.84]
Quintile	Name	VW	0.26 [3.34]	0.26 [3.22]	0.38 [5.42]	0.33 [4.73]	0.30 [4.26]	0.27 [3.78]
Quintile	Cap	VW	0.19 [2.56]	0.19 [2.56]	0.30 [4.62]	0.27 [4.04]	0.25 [3.68]	0.22 [3.28]
Decile	NYSE	VW	0.12 [1.35]	0.11 [1.16]	0.22 [2.60]	0.21 [2.50]	0.21 [2.41]	0.20 [2.34]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.22 [2.73]	0.21 [2.63]	0.32 [4.49]	0.30 [4.22]	0.24 [3.42]	0.22 [3.24]
Quintile	NYSE	EW	0.02 [0.35]					
Quintile	Name	VW	0.24 [3.01]	0.24 [2.94]	0.34 [4.86]	0.32 [4.51]	0.26 [3.80]	0.24 [3.55]
Quintile	Cap	VW	0.16 [2.21]	0.16 [2.11]	0.26 [3.91]	0.24 [3.60]	0.20 [3.01]	0.19 [2.80]
Decile	NYSE	VW	0.09 [0.95]	0.07 [0.76]	0.17 [2.01]	0.17 [1.97]	0.14 [1.68]	0.14 [1.65]

Table 3: Conditional sort on size and RAB

This table presents results for conditional double sorts on size and RAB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on RAB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high RAB and short stocks with low RAB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	RAB Quintiles					RAB Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.65 [2.66]	0.51 [1.97]	0.83 [3.26]	0.85 [3.38]	0.86 [3.07]	0.20 [1.80]	0.15 [1.33]	0.11 [0.95]	0.09 [0.75]	0.06 [0.51]	0.05 [0.39]
	(2)	0.74 [3.16]	0.74 [3.12]	0.74 [3.10]	0.72 [3.03]	0.92 [3.78]	0.18 [2.17]	0.15 [1.80]	0.13 [1.54]	0.17 [2.06]	0.09 [1.01]	0.13 [1.49]
	(3)	0.70 [3.33]	0.67 [3.14]	0.76 [3.50]	0.88 [3.92]	0.95 [4.19]	0.25 [2.85]	0.20 [2.27]	0.19 [2.19]	0.18 [1.96]	0.17 [1.90]	0.16 [1.73]
	(4)	0.62 [3.10]	0.56 [2.84]	0.72 [3.57]	0.87 [4.28]	0.79 [3.83]	0.17 [1.99]	0.15 [1.75]	0.22 [2.59]	0.21 [2.41]	0.15 [1.77]	0.15 [1.72]
	(5)	0.52 [3.01]	0.57 [3.23]	0.52 [3.24]	0.56 [3.36]	0.71 [4.22]	0.19 [2.17]	0.20 [2.26]	0.32 [4.15]	0.29 [3.59]	0.24 [3.09]	0.22 [2.72]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	RAB Quintiles					RAB Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	377	385	386	386	378	29	31	33	32	29	
	(2)	103	103	104	103	103	53	54	54	54	54	
	(3)	72	72	72	72	72	92	90	92	92	94	
	(4)	60	59	60	59	60	199	193	196	197	204	
(5)	54	54	54	54	54	1344	1401	1461	1788	1611		

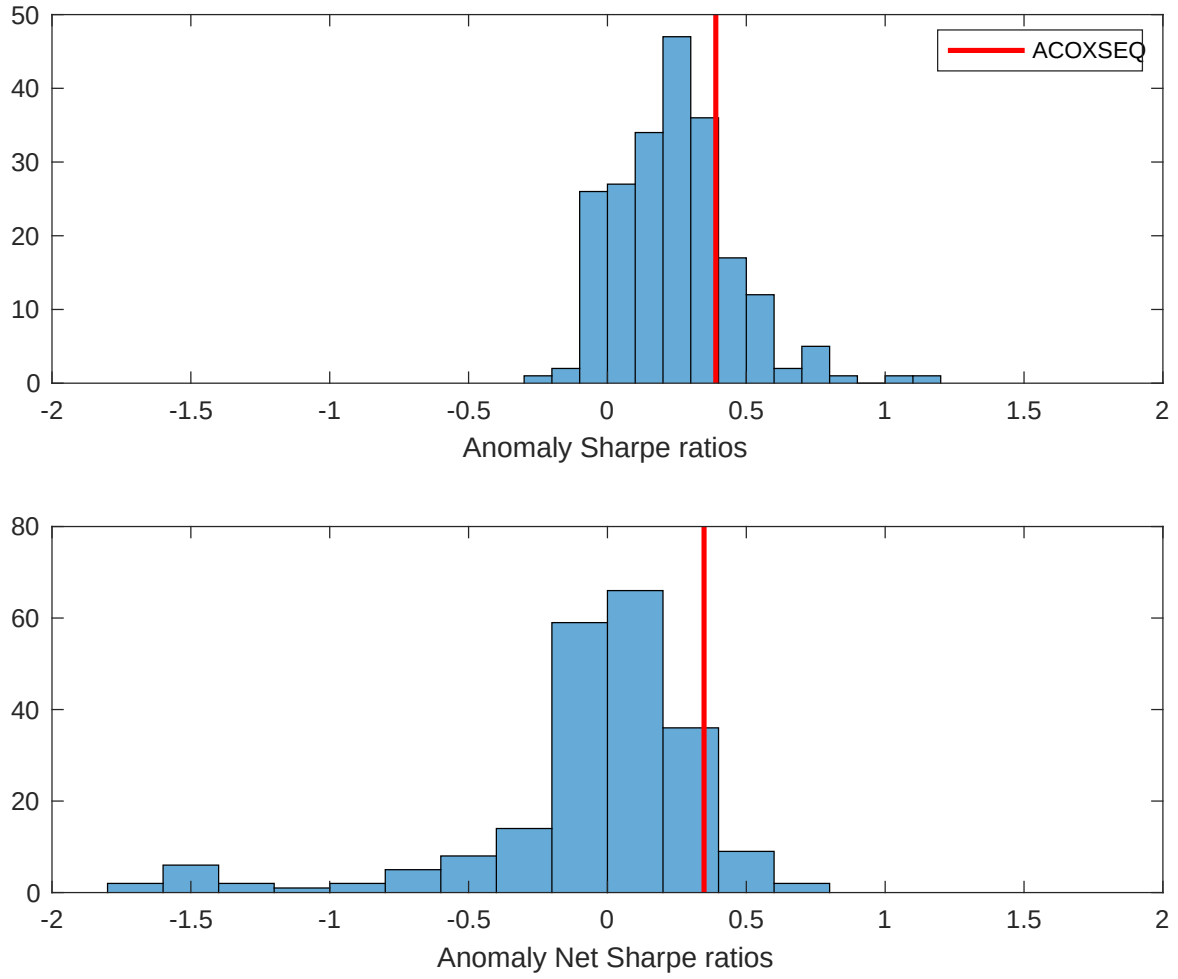


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the RAB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

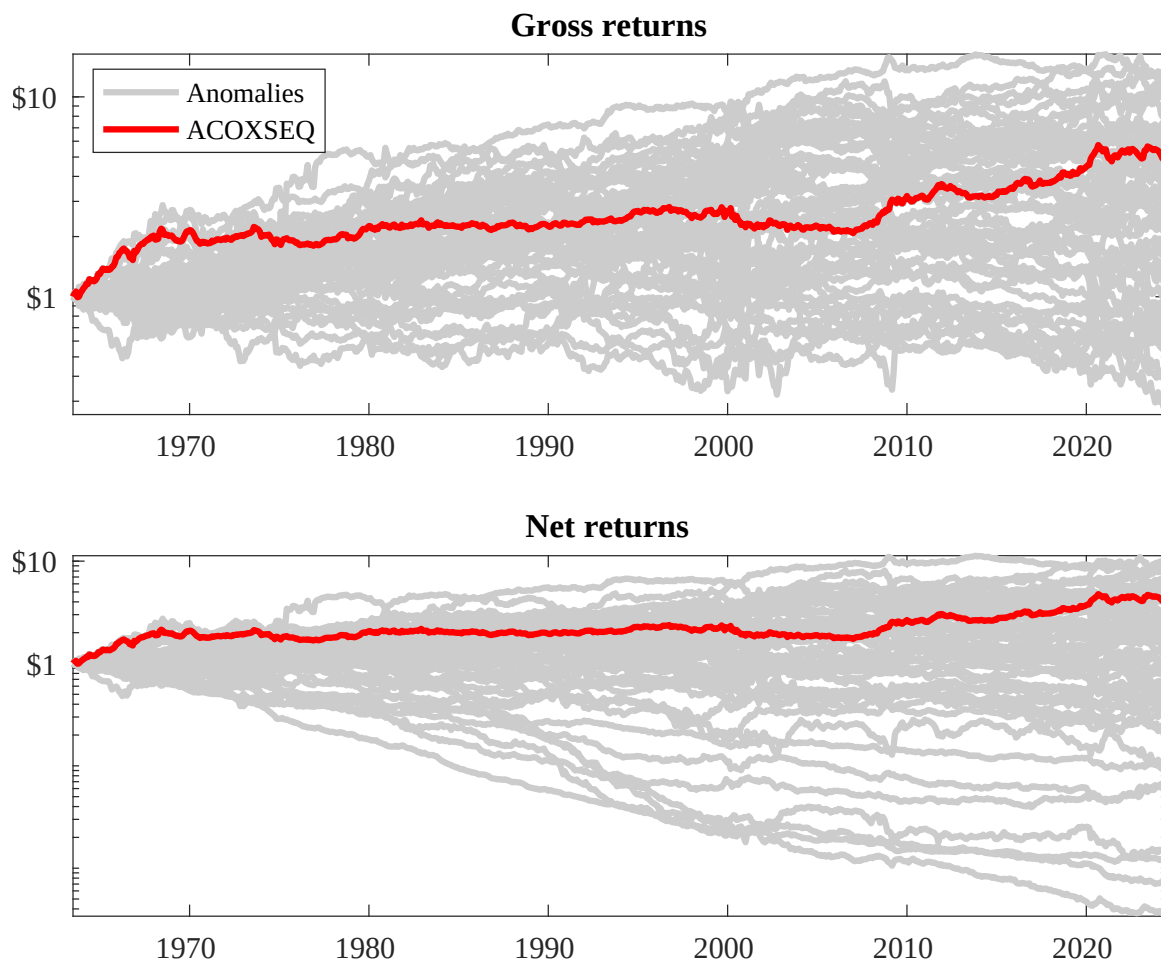
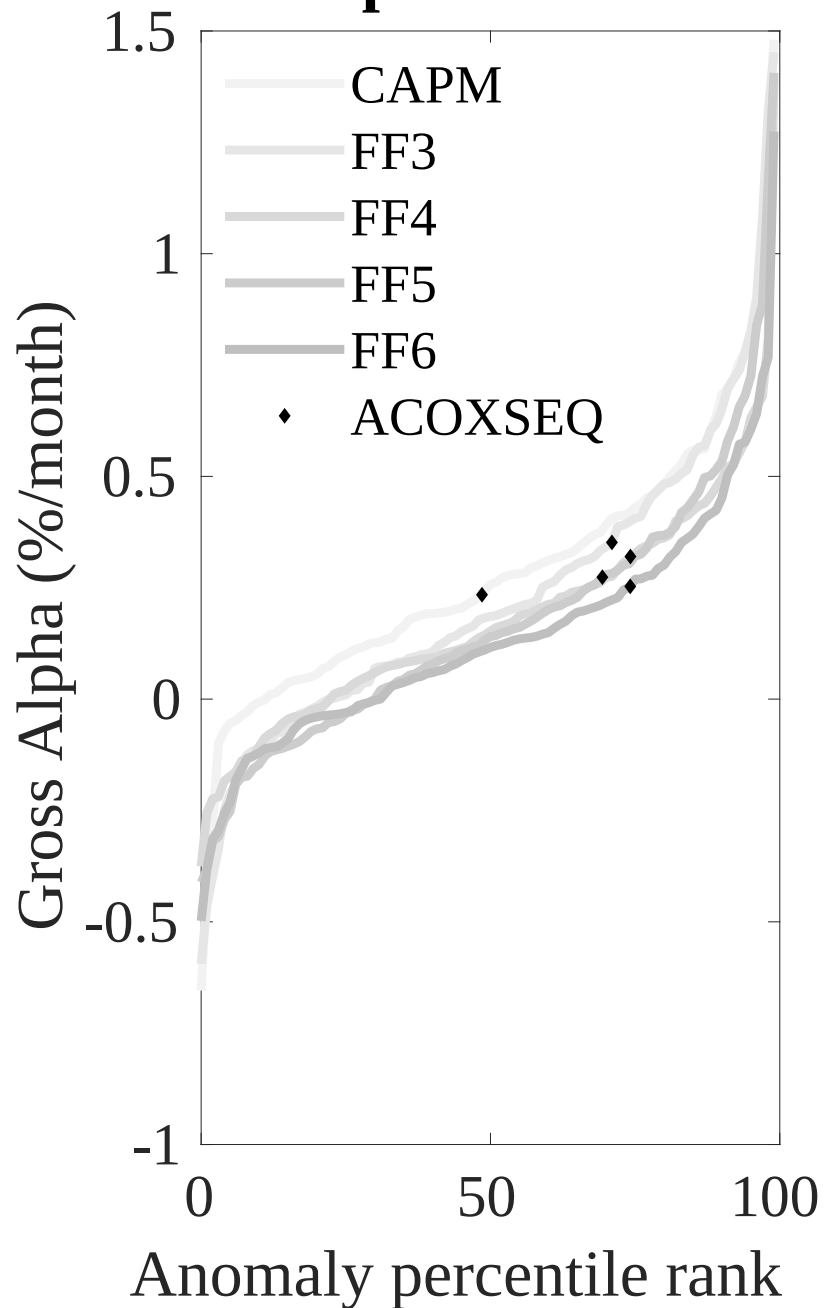


Figure 3: Dollar invested.

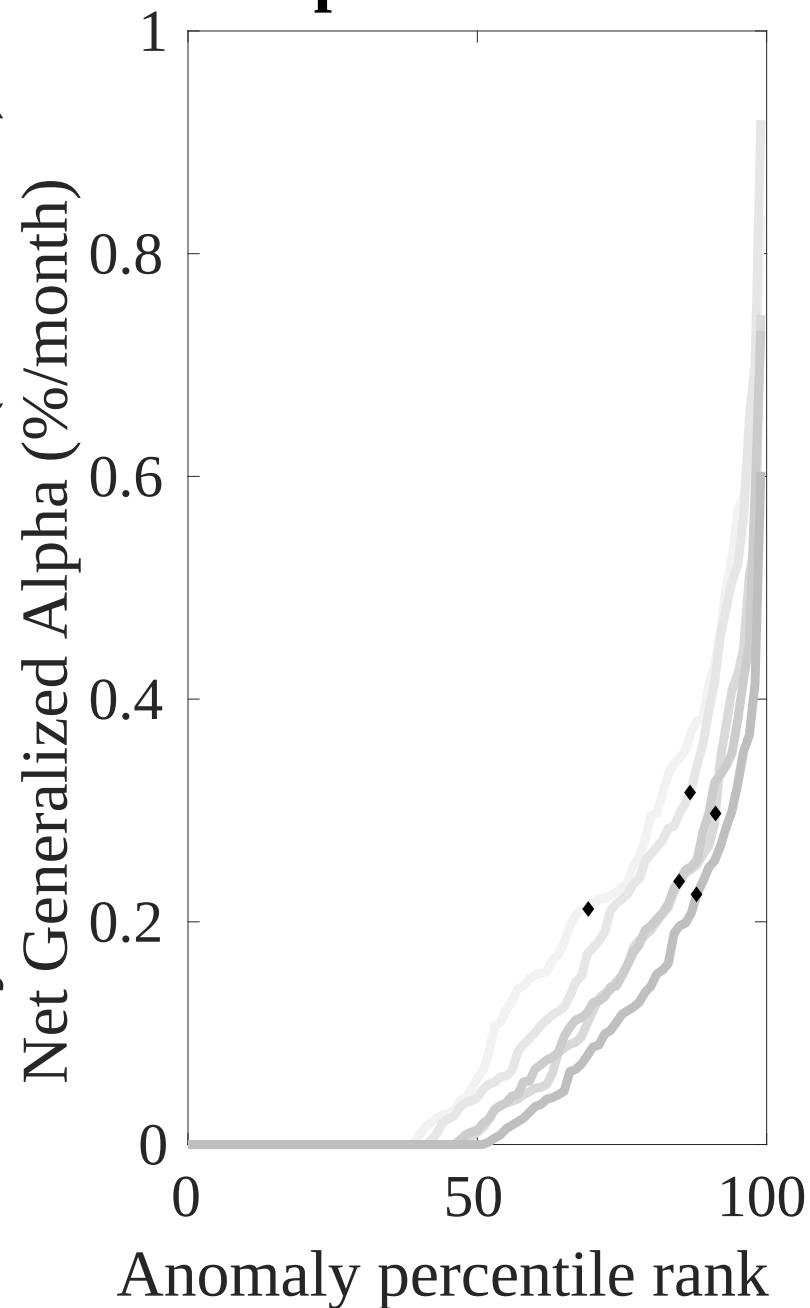
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the RAB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



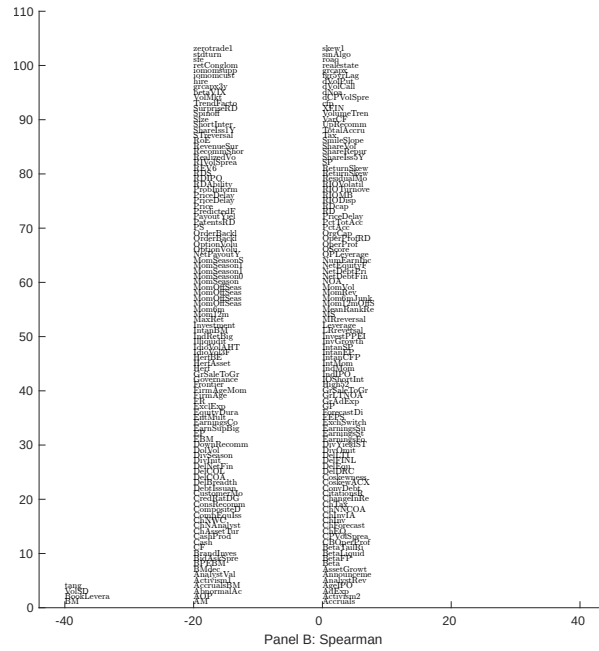
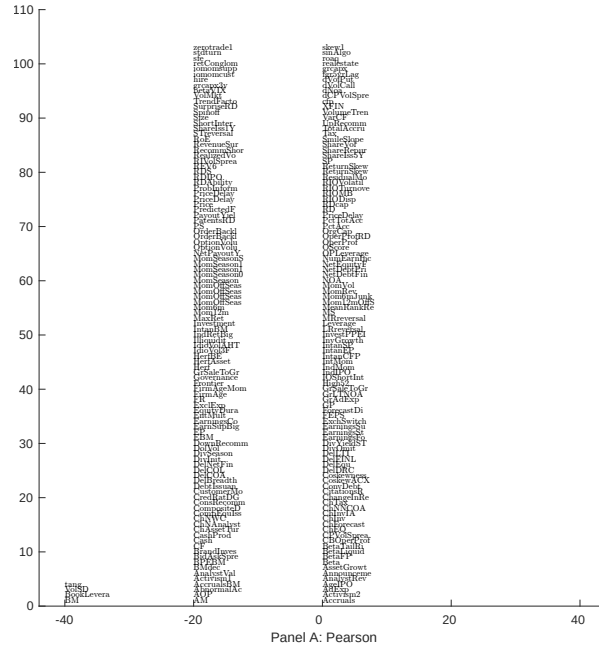


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with RAB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

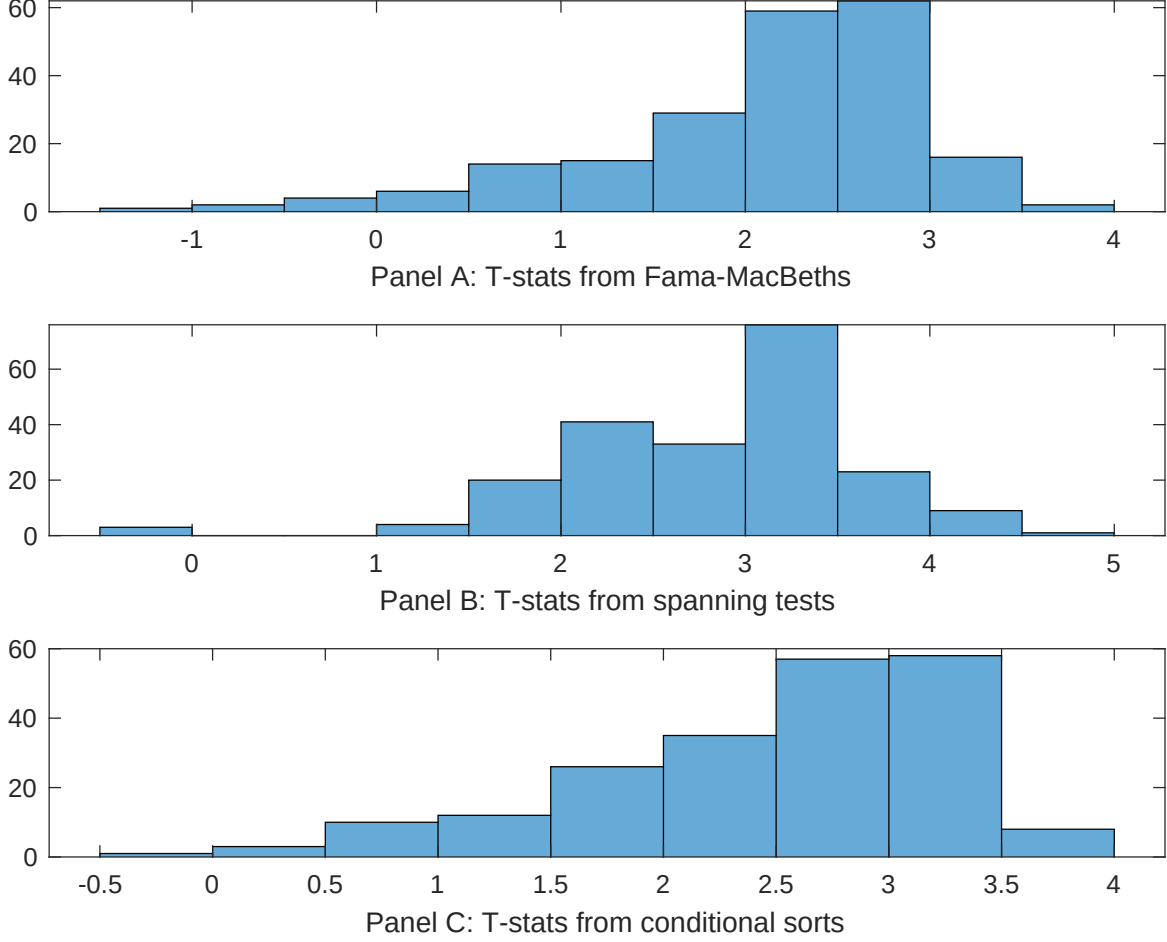


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of RAB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RAB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RAB}RAB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RAB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on RAB. Stocks are finally grouped into five RAB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RAB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on RAB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{RAB}RAB_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Book to market, original (Stattman 1980), Book leverage (annual), Operating leverage, Equity Duration, Earnings-to-Price Ratio, Enterprise Multiple. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [6.18]	0.12 [5.06]	0.10 [4.34]	0.20 [9.00]	0.11 [4.83]	0.13 [6.17]	0.21 [8.49]
RAB	0.13 [3.43]	0.11 [3.34]	0.74 [2.15]	0.12 [3.51]	0.81 [2.46]	0.87 [2.53]	0.90 [2.72]
Anomaly 1	0.37 [7.52]						0.87 [1.95]
Anomaly 2		0.13 [1.27]					-0.18 [-1.23]
Anomaly 3			0.12 [3.31]				0.88 [2.69]
Anomaly 4				0.51 [5.30]			0.60 [5.61]
Anomaly 5					0.17 [2.57]		-0.59 [-0.84]
Anomaly 6						0.79 [3.41]	0.10 [3.21]
# months	715	720	720	720	715	715	715
$\bar{R}^2(\%)$	1	0	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the RAB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{RAB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Book to market, original (Statman 1980), Book leverage (annual), Operating leverage, Equity Duration, Earnings-to-Price Ratio, Enterprise Multiple. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.23 [3.27]	0.22 [3.15]	0.23 [3.27]	0.25 [3.47]	0.26 [3.60]	0.30 [4.14]	0.21 [3.02]
Anomaly 1	-23.37 [-5.76]						-19.40 [-4.19]
Anomaly 2		10.55 [4.02]					7.48 [2.67]
Anomaly 3			19.47 [7.03]				16.55 [5.73]
Anomaly 4				-11.77 [-3.23]			-2.92 [-0.66]
Anomaly 5					-5.53 [-1.76]		11.04 [2.78]
Anomaly 6						-11.31 [-3.91]	-9.06 [-2.64]
mkt	-2.75 [-1.64]	-2.24 [-1.30]	-3.40 [-2.06]	-3.45 [-2.04]	-3.75 [-2.19]	-4.00 [-2.36]	-1.62 [-0.97]
smb	25.75 [9.06]	16.46 [6.75]	8.35 [3.11]	20.48 [7.70]	18.23 [7.16]	18.98 [7.62]	16.52 [5.35]
hml	-14.38 [-2.75]	-31.20 [-8.41]	-28.09 [-8.02]	-25.17 [-4.82]	-32.87 [-7.40]	-30.24 [-7.88]	-5.49 [-0.96]
rmw	15.25 [4.60]	22.01 [6.51]	8.09 [2.28]	18.52 [5.61]	19.67 [5.85]	19.95 [6.03]	8.15 [2.10]
cma	8.22 [1.73]	6.35 [1.33]	2.57 [0.54]	4.00 [0.83]	5.79 [1.19]	6.91 [1.44]	5.79 [1.23]
umd	4.22 [2.54]	2.67 [1.59]	2.74 [1.68]	3.00 [1.79]	2.20 [1.24]	-0.86 [-0.44]	2.07 [1.03]
# months	728	732	732	732	728	728	728
$\bar{R}^2(\%)$	31	30	33	29	28	30	36

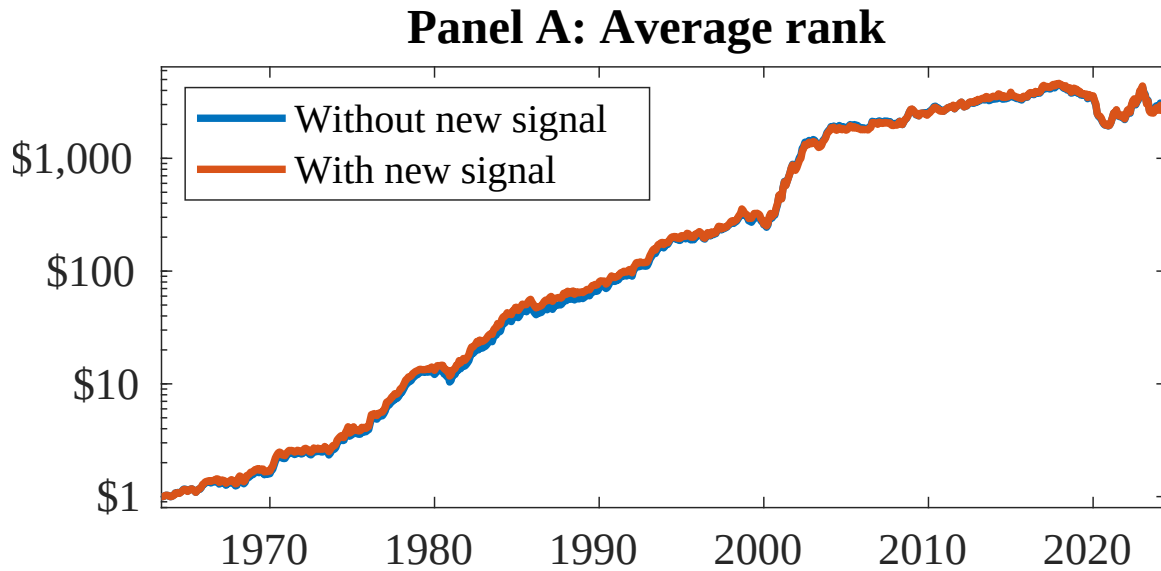


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as RAB. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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